

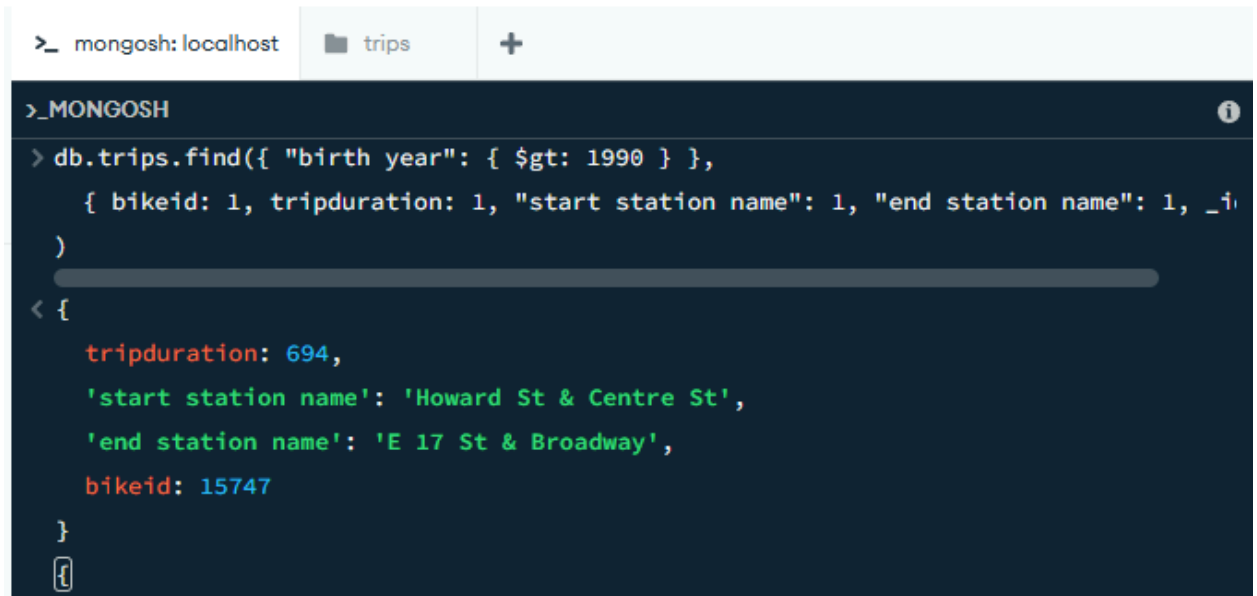
1. Recupera los viajes realizados con la bicicleta con ID 14785.

```
>_ mongosh: localhost  trips  +
>_MONGOSH
> db.trips.find({ bikeid: 14785})
< {
  _id: ObjectId('572bb8222b288919b68abf82'),
  tripduration: 812,
  'start station id': 518,
  'start station name': 'E 39 St & 2 Ave',
  'end station id': 433,
  'end station name': 'E 13 St & Avenue A',
  bikeid: 14785,
  usertype: 'Subscriber',
  'birth year': 1977,
  'start station location': {
    type: 'Point',
    coordinates: [
      -73.9734419,
      40.74780373
    ]
  }
},
```

2. Encuentra los viajes que duraron más de 1 hora.

```
>_ mongosh: localhost  trips  +
>_MONGOSH
> db.trips.find({ tripduration: { $gt: 3600 }})
< {
  _id: ObjectId('572bb8222b288919b68ac07c'),
  tripduration: 326222,
  'start station id': 391,
  'start station name': 'Clark St & Henry St',
  'end station id': 310,
  'end station name': 'State St & Smith St',
  bikeid: 18591,
```

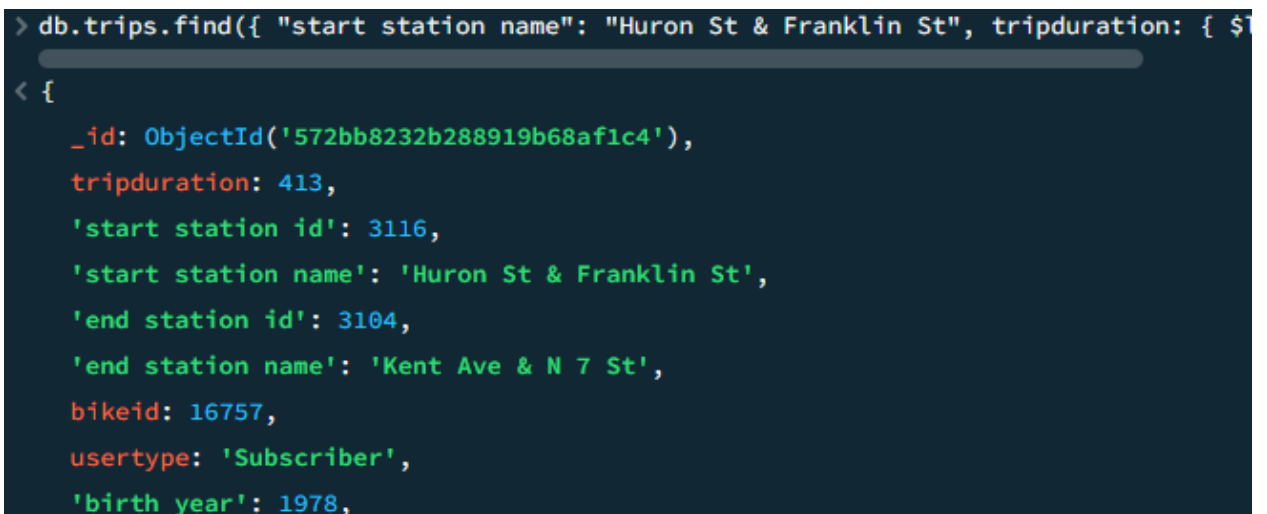
3. Mostrar solo el ID de la bicicleta, la duración del viaje y los nombres de las estaciones de los viajes realizados por usuarios nacidos después de 1990.



The screenshot shows a MongoDB shell window with the following content:

```
> mongosh: localhost trips +
>_MONGOSH
> db.trips.find({ "birth year": { $gt: 1990 } },
  { bikeid: 1, tripduration: 1, "start station name": 1, "end station name": 1, "_id": 0 })
< {
  tripduration: 694,
  'start station name': 'Howard St & Centre St',
  'end station name': 'E 17 St & Broadway',
  bikeid: 15747
}
```

4. Recupera los viajes realizados desde la estación Huron St & Franklin St y con duración menor a 500 segundos.



The screenshot shows a MongoDB shell window with the following content:

```
> db.trips.find({ "start station name": "Huron St & Franklin St", tripduration: { $lt: 500 } })
< {
  _id: ObjectId('572bb8232b288919b68af1c4'),
  tripduration: 413,
  'start station id': 3116,
  'start station name': 'Huron St & Franklin St',
  'end station id': 3104,
  'end station name': 'Kent Ave & N 7 St',
  bikeid: 16757,
  usertype: 'Subscriber',
  'birth year': 1978,
}
```

5. Encontrar los 10 viajes más largos, ordenados de mayor a menor duración.

```
>_ mongosh: localhost trips +
>_MONGOSH
> db.trips.find().sort({ duration: -1 }).limit(10)
< {
  _id: ObjectId('572bb8222b288919b68abf62'),
  tripduration: 1480,
  'start station id': 2006,
  'start station name': 'Central Park S & 6 Ave',
  'end station id': 2006,
  'end station name': 'Central Park S & 6 Ave',
  bikeid: 14562,
  usertype: 'Subscriber',
  'birth year': 1952,
  'start station location': {
    type: 'Point',
    coordinates: [
```

6. Encontrar viajes con estaciones que contengan Ave en su nombre.

```
>_ mongosh: localhost trips +
>_MONGOSH
> db.trips.find({"start station name": { $regex: "Ave"}})
< {
  _id: ObjectId('572bb8222b288919b68abf5a'),
  tripduration: 379,
  'start station id': 476,
  'start station name': 'E 31 St & 3 Ave',
  'end station id': 498,
  'end station name': 'Broadway & W 32 St',
  bikeid: 17827,
  usertype: 'Subscriber',
```

7. ¿Cuántos viajes han realizado los usuarios de tipo Customer?

```
>_ mongosh: localhost | trips | +
>_MONGOSH
> db.trips.countDocuments({ usertype: "Customer"})
< 1989
sample_training>
```

8. Recuperar los viajes realizados entre el 1 de enero y el 31 de marzo de 2016.

```
>_ mongosh: localhost | trips | +
>_MONGOSH
> db.trips.find({ "start time":
  { $gte: new Date("2016-01-01"), $lte: new Date("2016-03-31")})
< {
  _id: ObjectId('572bb8222b288919b68abf5a'),
  tripduration: 379,
  'start station id': 476,
  'start station name': 'E 31 St & 3 Ave',
  'end station id': 498,
  'end station name': 'Broadway & W 32 St',
```

9. Encontrar viajes con duración entre 10 y 20 minutos, hechos por usuarios nacidos antes de 1980.

```
>_ mongosh: localhost trips +
>_MONGOSH
> db.trips.find({ tripduration: { $gte: 600, $lte: 1200 },
  "birth year": { $lt: 1980 } })
< {
  _id: ObjectId('572bb8222b288919b68abf5b'),
  tripduration: 889,
  'start station id': 268,
  'start station name': 'Howard St & Centre St',
  'end station id': 3002,
  'end station name': 'South End Ave & Liberty St',
  bikeid: 22794,
  usertype: 'Subscriber',
  'birth year': 1961,
  'start station location': {
```

10. Encontrar los viajes que duraron más de una hora o que fueron realizados por usuarios nacidos después del año 2000

```
>_ mongosh: localhost trips +
>_MONGOSH
> db.trips.find({ $or: [{ tripduration: { $gt: 3600 } },
  { "birth year": { $gt: 2000 } } ] })
< {
  _id: ObjectId('572bb8222b288919b68ac07c'),
  tripduration: 326222,
  'start station id': 391,
  'start station name': 'Clark St & Henry St',
  'end station id': 310,
  'end station name': 'State St & Smith St',
  bikeid: 18591,
  usertype: 'Subscriber',
  'birth year': 1979,
  'start station location': {
    type: 'Point',
    coordinates: [
      -73.99344559,
      40.69760127
```

11. Mostrar los 5 viajes más recientes, pero saltando los primeros 10 resultados.

```
>_ mongosh: localhost | trips | +
>_MONGOSH
> db.trips.find().sort({ "start time": -1}).skip(10).limit(5)
< {
  _id: ObjectId('572bb8232b288919b68b1c29'),
  tripduration: 700,
  'start station id': 545,
  'start station name': 'E 23 St & 1 Ave',
  'end station id': 229,
  'end station name': 'Great Jones St',
  bikeid: 23922,
  usertype: 'Subscriber',
  'birth year': 1987,
  'start station location': {
```